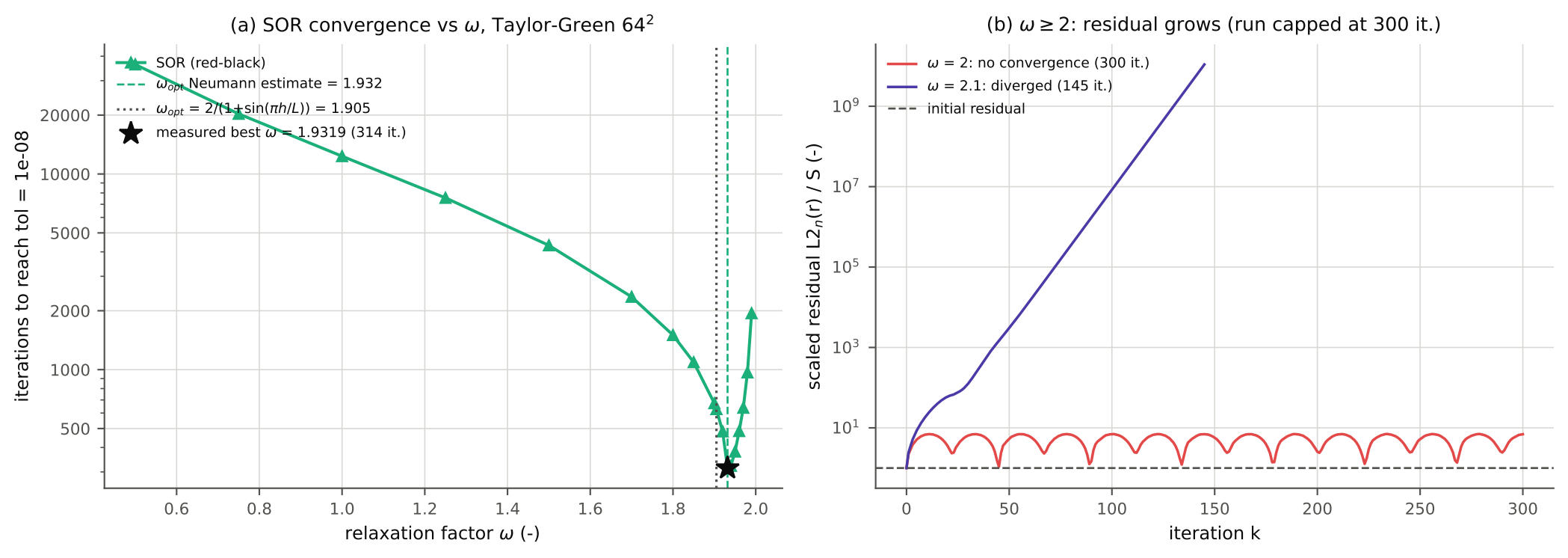
(a) SOR convergence vs ω, Taylor-Green 64²
SOR (red-black)
ω_opt Neumann estimate = 1.932
ω_opt = 2/(1+sin(πh/L)) = 1.905
measured best ω = 1.9319 (314 it.)
iterations to reach tol = 1e-08
20000
10000
5000
2000
1000
500
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
relaxation factor ω (-)
(b) ω ≥ 2: residual grows (run capped at 300 it.)
ω = 2: no convergence (300 it.)
ω = 2.1: diverged (145 it.)
initial residual
scaled residual L2_n(r) / S (-)
10⁹
10⁷
10⁵
10³
10¹
0
50
100
150
200
250
300
iteration k